

SPONDIN AGCM, AB, Canada

WMO: 714130

Lat: 51.816N

Lon: 111.683W

Elev: 804

StdP: 92.03

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.9	-28.9	-35.4	0.1	-31.8	-32.2	0.2	-28.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	30.1	16.4	28.0	16.0	26.0	15.2	18.4	25.1	17.3	24.4	16.5	23.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.6	21.0	14.9	11.7	20.1	13.8	10.9	18.7	55.3	25.2	51.9	24.2	49.1	23.7	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	DB	-35.8	33.3	2.3	2.3	-37.5	35.0	-38.8	36.4	-40.1	37.7	-41.8	39.4
			WB	-36.2	20.6	2.3	1.2	-37.8	21.5	-39.2	22.2	-40.5	22.9	-42.2	23.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-10.6	-12.3	-4.9	2.8	9.9	14.3	17.2	16.3	11.1	3.6	-4.8	-11.9	(6)
(7)		DBStd	12.29	8.31	8.17	7.99	5.26	4.27	2.95	2.78	3.12	4.57	4.92	7.28	8.16	(7)
(8)		HDD10.0	3371	639	623	462	221	55	3	0	1	40	203	444	678	(8)
(9)		HDD18.3	5766	897	857	721	467	261	126	55	77	219	456	694	937	(9)
(10)		CDD10.0	687	0	0	0	3	53	131	223	198	73	6	0	0	(10)
(11)		CDD18.3	41	0	0	0	0	1	3	19	15	2	0	0	0	(11)
(12)		CDH23.3	1167	0	0	0	4	59	109	461	390	143	1	0	0	(12)
(13)		CDH26.7	332	0	0	0	0	7	19	144	118	44	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	355	16	10	15	22	40	73	59	48	28	14	14	12	(15)
(16)		PrecMax	465	42	23	36	51	83	153	148	87	74	34	53	19	(16)
(17)		PrecMin	264	3	1	4	3	4	14	23	11	3	5	1	2	(17)
(18)		PrecStd	66	9	5	8	13	24	32	27	25	18	7	12	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	6.8	14.6	21.7	27.5	29.1	33.1	32.7	30.6	21.9	14.8	7.1	(19)
(20)			MCWB	3.2	3.4	7.4	9.6	13.5	15.8	17.5	15.8	15.3	11.3	7.5	3.3	(20)
(21)		2%	DB	4.7	3.2	11.2	18.7	24.9	26.2	30.2	29.7	27.7	18.2	10.2	4.1	(21)
(22)			MCWB	2.0	0.7	5.3	8.8	12.3	15.2	17.2	16.1	14.6	9.9	5.2	1.2	(22)
(23)		5%	DB	2.9	1.5	7.9	15.4	22.6	24.1	28.0	27.4	24.1	15.5	6.8	1.7	(23)
(24)			MCWB	0.7	-0.2	3.7	6.7	11.7	14.9	16.8	15.9	13.5	8.3	3.0	-0.6	(24)
(25)		10%	DB	1.0	-0.4	4.9	12.4	19.8	22.2	25.7	25.4	20.9	12.3	4.2	-0.9	(25)
(26)			MCWB	-0.7	-1.6	1.9	5.3	10.4	14.1	16.1	15.5	12.0	6.7	1.5	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.0	8.1	11.1	14.5	18.3	20.9	19.2	16.6	12.7	8.2	3.8	(27)
(28)			MCDWB	7.1	6.4	14.5	20.6	24.6	24.8	28.6	25.9	27.2	19.8	14.8	6.8	(28)
(29)		2%	WB	2.1	1.1	5.6	8.8	13.3	16.8	18.8	18.1	15.1	10.5	5.4	1.4	(29)
(30)			MCDWB	4.5	2.8	10.3	17.3	22.7	23.2	25.4	25.0	25.0	16.3	9.8	4.0	(30)
(31)		5%	WB	1.0	-0.1	3.7	7.2	12.3	15.7	17.7	17.0	13.9	8.7	3.4	-0.4	(31)
(32)			MCDWB	2.8	1.4	7.7	14.2	20.8	22.2	24.6	24.5	22.4	14.7	6.2	1.6	(32)
(33)		10%	WB	-0.6	-1.6	1.9	5.8	11.1	14.7	16.9	16.0	12.6	7.1	1.8	-2.4	(33)
(34)			MCDWB	0.9	-0.3	4.5	11.6	18.5	20.5	23.9	23.2	20.6	12.1	3.9	-0.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	11.0	11.1	13.0	14.4	13.1	14.5	14.8	15.0	13.0	10.7	10.2	(35)
(36)			MCDBR	10.8	11.7	14.4	19.4	19.5	17.0	19.4	19.7	21.3	18.0	14.5	12.4	(36)
(37)			MCWBR	8.6	9.7	9.0	10.1	8.7	7.6	8.1	8.0	9.7	10.2	9.6	9.9	(37)
(38)		5% WB	MCDBR	11.2	11.4	13.9	17.3	16.6	14.5	16.1	16.7	19.5	17.0	13.5	12.5	(38)
(39)			MCWBR	9.2	9.6	9.0	9.5	7.9	7.2	7.9	7.7	9.3	9.8	9.4	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.311	0.286	0.336	0.363	0.368	0.358	0.373	0.321	0.284	0.256	0.254	(40)	
(41)		taud	2.306	2.217	2.462	2.342	2.313	2.345	2.410	2.338	2.460	2.540	2.483	2.322	(41)	
(42)		Ebn at Noon	747	798	923	902	889	885	889	853	869	847	782	705	(42)	
(43)		Edh at Noon	62	91	87	111	121	118	109	111	87	66	52	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.14	2.19	3.59	4.81	5.81	6.05	6.46	5.25	3.74	2.25	1.18	0.86	(44)	
(45)		RadStd	0.10	0.18	0.28	0.41	0.39	0.38	0.29	0.39	0.36	0.28	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page